

Tutorial Sheet - Topic 07

Programming Fundamentals 1

Methods, Parameters, Return Values, and Drivers

Part A – Method Basics

Q1. Terminology (recap)

Label each part of the method below:

```
public void printHello() {  
    System.out.println("Hello");  
}
```

- (a) What is the **return type**?
 - (b) What is the **method name**?
 - (c) What are the **parameters**?
 - (d) What is inside the **method body**?
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Q2. Why use methods?

List **two reasons** why we use methods instead of repeating code.

Q3. Predict the output

What is printed when this code runs?

```
public void greet() {  
    System.out.println("Hi there!");  
    System.out.println("Welcome to Java!");  
}
```

If we call `greet()` twice, what happens?

Part B – Writing and Calling Methods

Q4. Writing your own method

Write a method called `sayName` that takes **no parameters** and prints your own name.

Q5. Adding parameters

Write a method called `printSum` that takes **two integers** as parameters and prints their sum.

Example:

If called as `printSum(4, 6);`

→ Output should be: `The sum is 10`

Q6. Return values

Write a method called `timesTwo` that takes one integer parameter and **returns** the number doubled.

Example:

If called as `timesTwo(8)` → the method should return `16`.

Q7. Using a driver class

Explain briefly what the **Driver** class does in Java.

Why is the `main()` method usually placed there?

Part C – Combining Methods and Drivers

Q8. Read the example and answer:

```
public class Calculator {
    public int add(int x, int y) {
        return x + y;
    }
}
```

```
public class CalculatorDriver {
    public static void main(String[] args) {
        Calculator calc = new Calculator();
        int result = calc.add(5, 3);
        System.out.println("Sum = " + result);
    }
}
```

- (a) What line **creates an object** of the Calculator class?
 - (b) What line **calls** the method?
 - (c) What is printed on screen?
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Q9. Extending the joke generator

Imagine a class `JokeGenerator` with three methods:

- `tellJoke()` – prints a joke (no input)
- `tellStory(String name)` – prints a story with a name
- `ageInTenYears(int age)` – returns a String message

Write a short pseudocode outline of a **Driver** class that:

1. Creates a `JokeGenerator` object
 2. Calls each of the three methods in turn
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Q10. Writing other methods with parameters.

10.1 Write a method called `multiply` that takes **two integers** as parameters and **returns** their product.

10.2 Write a method that calls `multiply` and prints the result.

10.3 Where is the `multiply` method called?

Q11 . Writing other methods with parameters.

11.1 Write a method called `add3` that takes **three integers** as parameters and **returns** their sum.

11.2 Write a method that calls `add3` and prints the result.

11.3 Where is the `add3` method called?

Part D – Beyond the Basics

Q12. Boolean method

Write a method `isEven(int number)` that returns `true` if the number is even, otherwise `false`.

Then write one line of code that prints "Even!" if the method returns true.

Q13. Overloading

The following two methods are in the same class:

```
void addNum(int x, int y) { ... }  
void addNum(double x, double y) { ... }
```

(a) What is this feature called?

(b) What would happen if we called `addNum(3.5, 2.5);` ?

Q14. Challenge – Fahrenheit to Celsius

Write a method that converts from Fahrenheit to Celsius and returns the Celsius value as a `double`.

Write the **method header** and one **example call** from `main()`.

Formula:

$$C = (F - 32) \times 5 / 9$$

Optional Discussion Questions

- Why might Boolean methods make your program clearer?
 - How do method parameters help avoid global variables?
 - What makes overloaded methods useful in real programs?
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